

Math AA HL at KCA - Chapters 6 to 8 Notes

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1 Quadratics

A quadratic function can be in any of the following three forms

Form	General expression	Features
Standard	$ax^2 + bx + c$	$c = y$ -intercept
Vertex	$a(x - h)^2 + k$	(h, k) vertex; $x = h$ line of symmetry
Factored	$a(x - \alpha)(x - \beta)$	α, β roots

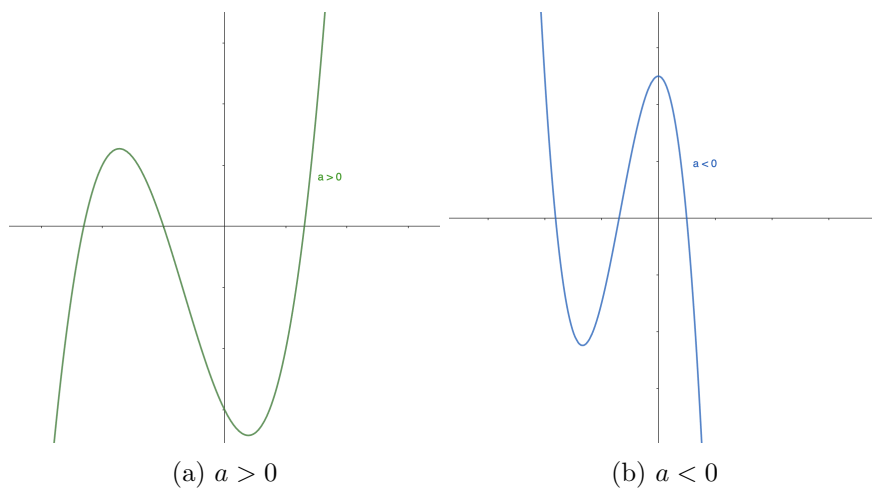
The leading coefficient a determines the concavity of the function

- $a > 0 \implies$ the function is convex (tends to ∞)
- $a < 0 \implies$ the function is concave (tends to $-\infty$)

2 Sketching Cubics and Higher Degree Graphs

A polynomial of degree n can have at most n distinct roots.

The following features are to be considered when sketching a polynomial $f(x)$ with leading coefficient a



- Behaviors as $x \rightarrow \pm\infty$; for a cubic:
 - $a > 0 \implies \text{as } x \rightarrow \pm\infty, y \rightarrow \pm\infty$
 - $a < 0 \implies \text{as } x \rightarrow \pm\infty, y \rightarrow \mp\infty$
- If the roots are α, β, γ , then the x -intercepts are $(\alpha, 0)$, $(\beta, 0)$, $(\gamma, 0)$.
- The y -intercept, this can be found by $f(0)$
- The turning point(s), if applicable and necessary.

3 Mappings

A **relation** describes how values are mapped from one set onto another.

In the case of functions, values from the **domain** are mapped onto values in the **range**.

There are four types of mapping

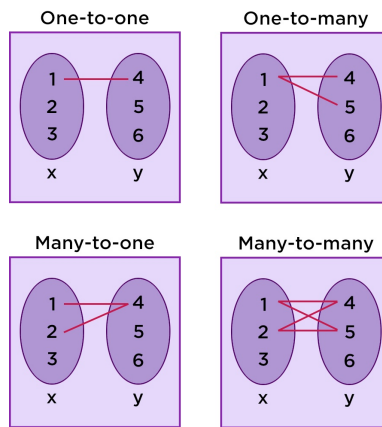


Figure 2: Mappings

For a relation to be a function, it has to be either one-to-one or many-to-one. This is because each x value inputted into the function must be associated with a single y value.

3.1 Interval Notations

An opening/closing square bracket represents an inclusive end.

An opening/closing parenthesis represents an exclusive end.

E.g.

- $x \in [a, b) \iff a \leq x < b$
- $x \in (-\infty, h] \iff x \leq h$

3.2 Domain and Range

The **range** is the set of x values that can go into a function, and the **range** is the set of y values that can come out of a function.

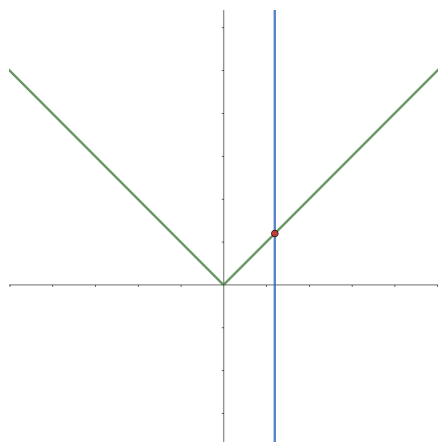
Notations for expressing a domain

1. $x \in [a, b]$
2. $a \leq x \leq b$
3. $R = \{x \mid a \leq x \leq b\}$

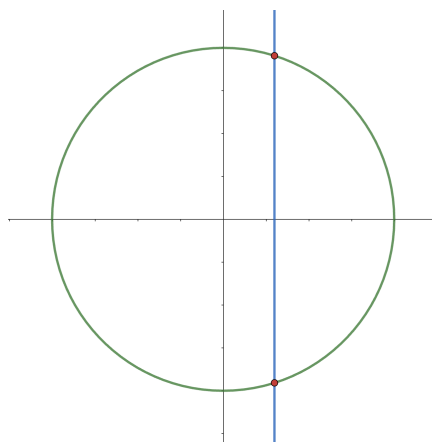
Notations for expressing a range

1. $y \in [a, b]$
2. $f(x) \in [a, b]$
3. $a \leq f(x) \leq b$
4. $D = \{y \mid a \leq y \leq b\}$

3.3 Restricting the Domain



(a) Function



(b) Not a function

A relation is not a function if it does not pass the **vertical line test**, which states that: If *at any point* in the domain of a relation, a vertical line intersects the graph at two or more points, then it is not a function.

For such a relation to be a function, the domain needs to be restricted such that the new domain passes the vertical line test.

4 Composite Functions

$$(f \circ g)(x) = f(g(x))$$

Properties

- Associativity $f \circ (g \circ h) = (f \circ g) \circ h$
- $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$

5 Inverse Functions

The inverse function of a function $f(x)$ is denoted $f^{-1}(x)$, it is a reflection in the line $y = x$.

A **self-inverse** function is its own inverse, i.e. $f(x) \equiv f^{-1}(x)$. E.g. $y = \frac{1}{x}$ is a self-inverse function.

Implications:

- $(f^{-1} \circ f)(x) = x$ and $(f \circ f^{-1})(x) = x$
- The *domain* of f is the *range* of f^{-1} , and vice versa
- The intersections of $f(x)$ and $f^{-1}(x)$ lie on the line $y = x$ and can be found by $f(x) = x$ or $f^{-1}(x) = x$

To find an expression for the inverse function, exchange the x 's and y 's, then make the new y the subject.

A function has an inverse if and only if it is **injective** over its domain, i.e. it is a one-to-one mapping over its domain. This can be examined using the *horizontal line test*.

Sometimes to restrict the domain such that there is an inverse, the turning point is used.

6 Even and Odd Functions

1. Even functions: $f(-x) = f(x)$; symmetrical about the y -axis
 - (a) $f(x)$ has only even powers of x
2. Odd functions: $f(-x) = -f(x)$; rotational symmetry of 180° about the origin
 - (a) $f(x)$ has only odd powers of x

7 Function Transformations and Descriptions

$-f(x)$	Reflection in the x -axis
$f(-x)$	Reflection in the y -axis
$f(x + a)$	Translation to the left by a units
$f(x - a)$	Translation to the right by a units
$f(kx)$	Horizontal stretch by a factor of $\frac{1}{k}$
$f(x) + b$	Upward translation by b units
$f(x) - b$	Downward translation by b units
$kf(x)$	Vertical stretch by a factor of k

8 Special Functions and Their Graphs

Sketching these functions involves looking at

1. x and y intercepts
2. horizontal and vertical asymptotes
3. where $f(x)$ tends to as $x \rightarrow \pm\infty$
4. sometimes the turning points too

8.1 Rational/Reciprocal Functions

They take the general form of

$$f(x) = \frac{ax + b}{cx + d}$$

and have the following properties

- x -intercept = $(-\frac{b}{a}, 0)$; when $y = 0$ and $a \neq 0$
- y -intercept = $(0, \frac{b}{d})$; when $x = 0$ and $d \neq 0$
- vertical asymptote $x = -\frac{d}{c}$; when the denominator is 0
- horizontal asymptote $y = \frac{a}{c}$; cancel out the b , d , and x 's, as the limit is obtained for very large values of x .

8.1.1 Oblique Asymptotes

The oblique asymptotes can be found by dividing the numerator by the denominator, then the quotient is the oblique asymptote.

8.2 Exponential Functions

They can take the form of

$$f(x) = ka^{bx} + c, \quad a > 0$$

and have the properties

1. y -intercept $(0, k + c)$
2. Horizontal asymptote $y = c$
3. No vertical asymptote
4. $b > 0 \implies \lim_{x \rightarrow \infty} f(x) = \infty$; growth
5. $b < 0 \implies \lim_{x \rightarrow -\infty} f(x) = \infty$; decay

8.3 Logarithmic Functions

They are the reflections of exponential functions in $y = x$. They have vertical asymptotes but not horizontal asymptotes.

8.4 Modulus Functions

Arithmetic rules

1. $|-x| = |x|$
2. $|x|^2 = x^2$
3. $|xy| = |x||y|$
4. $\left|\frac{x}{y}\right| = \frac{|x|}{|y|}$

These functions are formally defined as

$$|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0 \end{cases}$$

The graph is obtained by reflecting all parts below the y -axis through the y -axis so that they are all above it now.

In general, there are three ways of solving modulus equations/inequalities; solving manually will require the results to be substituted back into the equation to validate the answers and discard any invalid ones

1. Split into cases. e.g. for $|a| = 3$, solve when $a = 3$ and also $a = -3$, then validate the results
2. If both sides are in absolute value, or only one side is and the other is a constant (e.g. $|a| = |b|$ or $|a| = 3$), then square both sides and solve from there
3. Sketch graphs and find intersections
4. Use the G.D.C

The transformation of absolute value functions can be *done step by step* and it is normally easier to visualize this way.

8.5 Graphs of $[f(x)]^2$

Feature of $f(x)$	Feature of $[f(x)]^2$
Parts where $y < 0$	$y > 0$
$(a, 0)$ is an x-intercept	$(a, 0)$ local minimum
$(0, b)$ is a y-intercept	$(0, b^2)$ y-intercept
$x = a$ is a vertical asymptote	unchanged
$y = b$ is a horizontal asymptote	$y = b^2$ becomes a horizontal asymptote
$y \rightarrow \pm\infty$	$y \rightarrow \infty$

8.6 Graphs of $\frac{1}{f(x)}$

Feature of $f(x)$	Feature of $\frac{1}{f(x)}$
$x = a$ is an x-intercept	$x = a$ is a vertical asymptote
$(0, b)$, $b \neq 0$ is a y-intercept	$(0, \frac{1}{b})$ becomes the y-intercept
$x = a$ is a vertical asymptote	x-intercept at $(a, 0)$
$y = b$ is a horizontal asymptote	$y = \frac{1}{b}$ is a horizontal asymptote
$y = 0$ is a horizontal asymptote	$y \rightarrow \pm\infty$
$y \rightarrow \pm\infty$	$y = 0$ is a horizontal asymptote
(a, b) , $b \neq 0$ is a turning point	$(a, \frac{1}{b})$ is a turning point